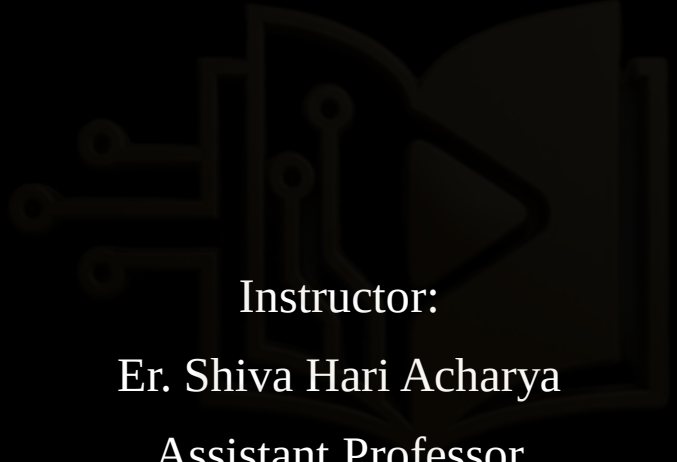


Chapter-1

INTRODUCTION TO INSTRUMENTATION SYSTEM



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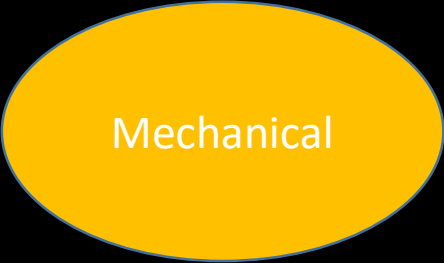
1.1 INTRODUCTION

1.1.1 Measurement:

- Measurement of a given quantity is the act of comparison between unknown and predefined standard.
- Result is expressed in numerical value.
- Physical parameters are converted into meaningful numbers.
- During measurement process:
 - The property of the object under consideration is compared to an accepted standard unit.
 - The number of times the unit standard fits into the quantity being measured gives the numerical value.
 - Numerical value is followed by a unit. It gains the unit of the standard.
- Numerical values are meaningless until it is followed by a unit.
- Methods of measurement may be direct or indirect.

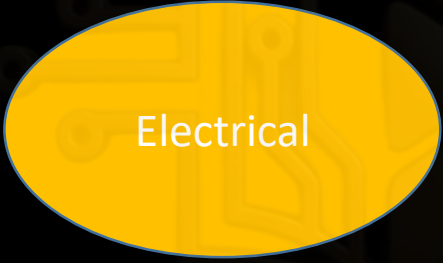
1.1.2 Instrument:

- Device that gives the measurement of any particular physical variable applied to it.
- The assembly of the instruments for measure, update and control the function is called Instrumentation System/ Measurement System.
- Instrument may be:



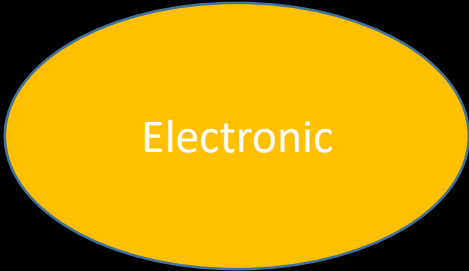
Mechanical

Eg: Strain gauge



Electrical

Eg: Galvanometer



Electronic

Eg: CRO

1.1.3 Accuracy:

- It is the closeness with which an instrument reading approaches the true value of the quantity being measured.
- Conformity to the truth.
- Accuracy may be specified in terms of inaccuracy or limits of error.

1.1.4 Precision:

- It is the measure of reproducibility of the measurement.
- It is the degree of agreement with in a group of measurements.
- Two parts: Conformity and Significant figure.
- Error caused due to the limitation of scale reading is called precision error.

1.1.5 Sensitivity:

- It is the measure of change in instrument output which occurs due to the change in quantity being measured.
- It is the ratio given by:

$$S = \frac{\text{Scale deflection}}{\text{value of measurand causing deflection}}$$

$$S = \text{Change in o/p} / \text{Change in i/p}$$

1.1.6 Resolution (Discrimination):

- Smallest increment in i/p which can be detected with certainty by an instrument is its resolution.
- If the i/p is slowly increased for some non zero i/p value, it will be found that o/p does not change until a certain increment is exceeded. This increment is called resolution.

1.1.7 Error:

- Difference between measured value and true value.
- Shows the deviation of measured value from the true value.
If measured value=9.8V and true value=10V
Error=9.8-10=-0.2V Here, -ve sign means measured value is less than true value.
- Types of error: Gross error, Systemic error and Residual error.

Assignment:

“ Precision is necessary but not sufficient condition for accuracy”. Justify your answer with suitable example.

1.2 Instrumentation system:

- Also known as Measurement system.
- The assembly of the instruments for measure, update and control the function is called Instrumentation System/ Measurement System.
- The block diagram consists of following major blocks:
 - Primary sensing element
 - Variable conversion element
 - Variable manipulation element
 - Data transmission element
 - Data Presentation element

Block diagram:

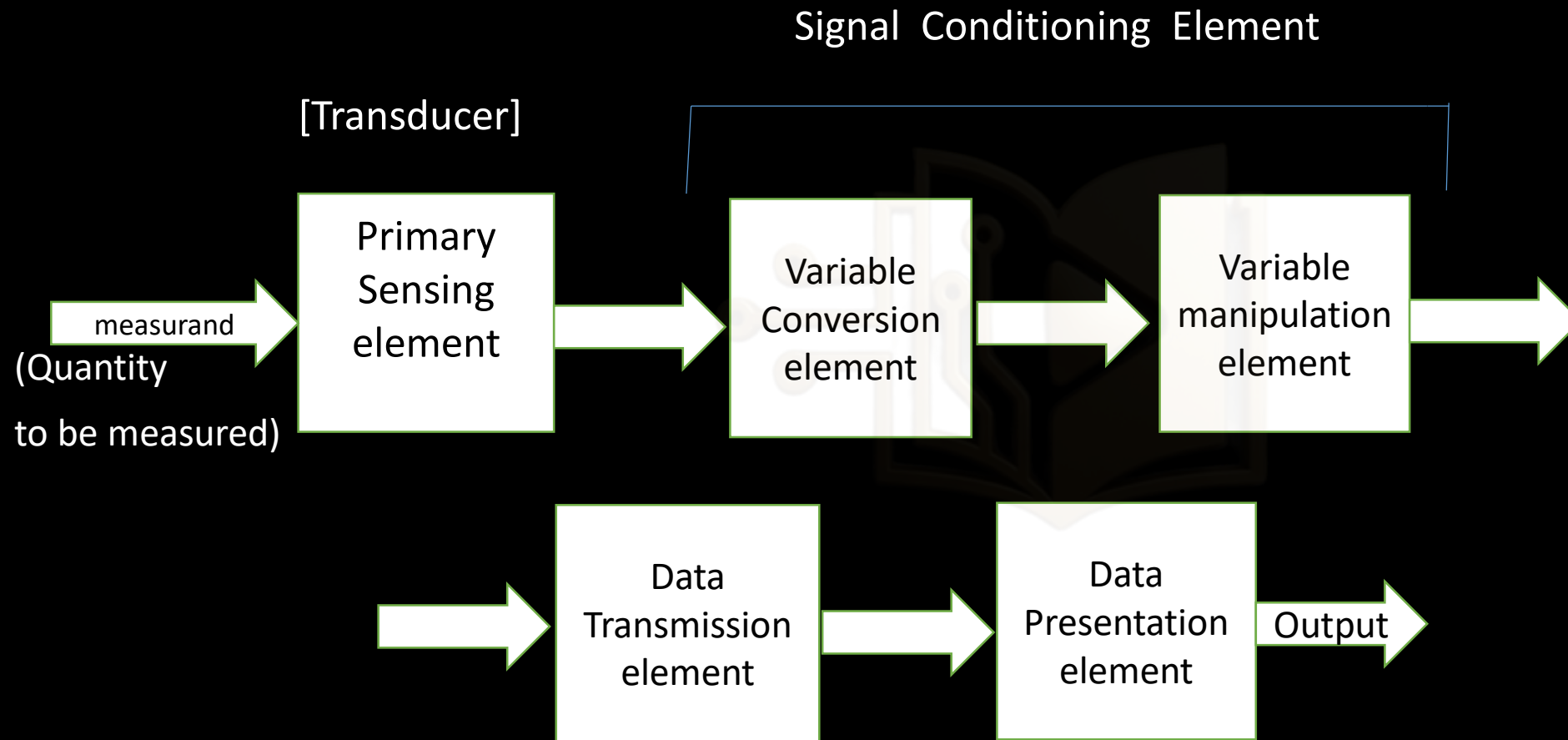


Fig: Block diagram of Instrumentation system

Primary sensing element:

- Measurand is first detected by primary sensing element.
- Transducer then converts it into electrical form.
- Transducer is a device which converts energy from one form to another.
- An electrical transducer converts any physical quantity into corresponding electrical quantity.
- The physical quantity to be measured is first sensed and detected by an element which gives the o/p in different analysis form. Then the o/p signal is converted into an electrical form by a transducer.
- Eg: microphone, loudspeaker, strain gauge.
- Detector-transducer stage.

Signal conditioning element:

a. Variable conversion element:

- O/p of transducer may be of an electrical signal of any form.
- It may be voltage, current, frequency or any other.
- May not be suited for system.
- To perform the desired function, it is necessary to convert the o/p in to suitable form.
- Eg: ADC, DAC.

b. Variable manipulation element:

- It manipulates the signal presented for preserving the original nature.
- Change in numerical value only.
- But preserves the original nature of the signal.
- Eg.: Amplifier.

Data transmission element:

- Instrumentation system consists of several instruments/ components which may be far apart.
- Communication/ data transmission is necessary between components or systems.
- The result is to be transmitted from a system to another system.
- This action is done with the help of data transmission element.
- Data transmission is done either thru wire (eg; twisted pair cable, coaxial cable, optical fibre or wave guide) or it may be wireless (eg; microwave, Bluetooth, infra red).
- Eg: Land line data transmission, RF data transmission.

Data presentation element:

- The information about the quantity under measurement has to be conveyed to the person who is handling the instrument.
- The final result/output is presented either thru visual devices, indicators or recorders.
- Eg: CRO, LED, strip chart recorder, magnetic tape recorder, X-Y recorder.
- Signal conditioning and transmission stage → Intermediate stage.
- Data presentation stage → Terminating stage.

1.3 Signal and its types:

1.3.1 Signal:

- A signal is a function representing a physical quantity, and typically it contains information about the behavior or nature of the phenomenon.
- A signal is any function that carries some information.

1.3.2 Types of signals:

- Continuous time and discrete time signals
- Periodic and aperiodic signals
- Deterministic and non deterministic signals
- Even and odd signals

Types of signals continue...

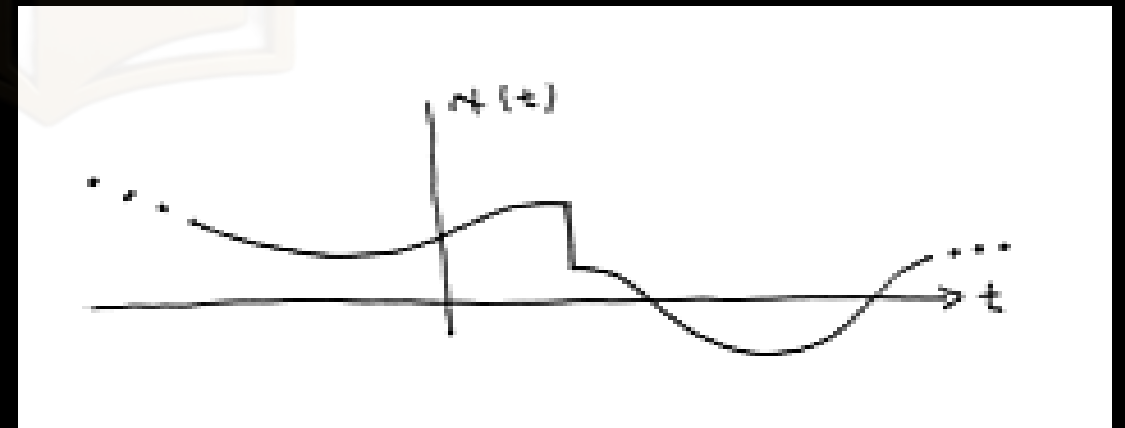
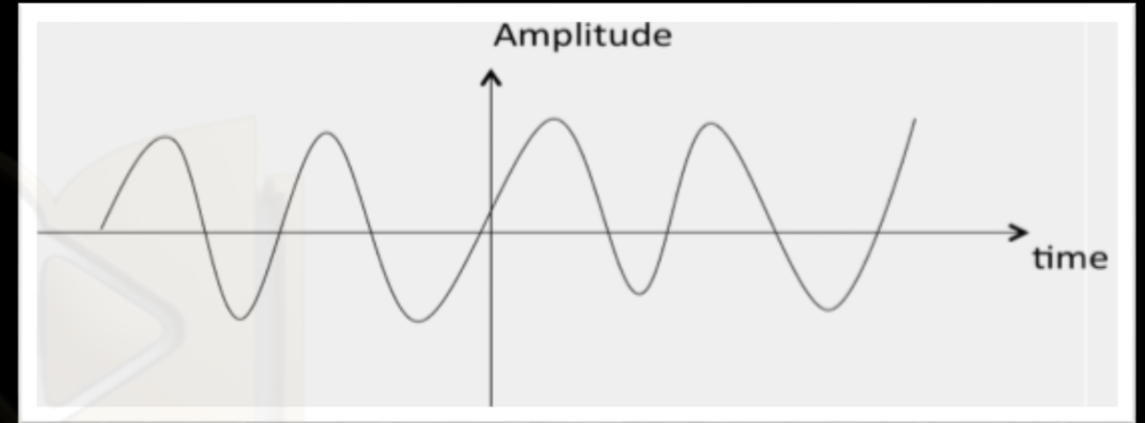
- Energy and power signals
- Unit step function
- Rectangular pulse
- Triangular pulse
- Sinc signal
- Impulse function/ Delta function
- Signum function



Continuous time and discrete time signal:

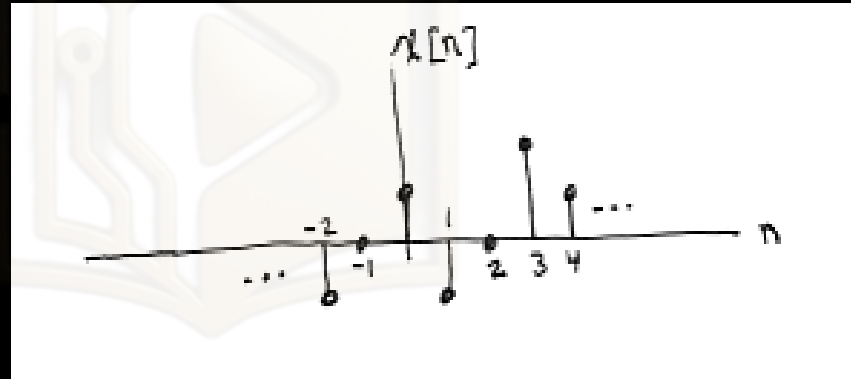
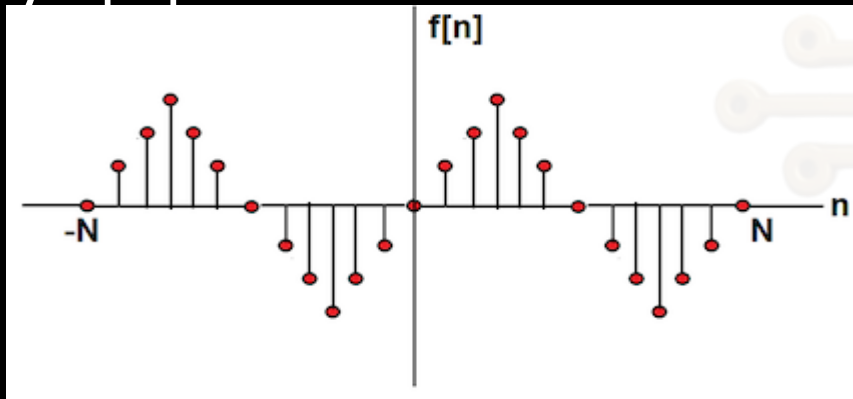
Continuous time signal:

- A signal is said to be continuous when it is defined for all instants of time.
- It is defined as a mathematical continuous function.
- Independent variable is continuous time 't' and the signal represented by $x(t)$.
- It is a function $x(t)$ of the real variable 't' defined for $-\infty < t < \infty$.



Discrete time signal:

- Defined only at certain time intervals.
- Amplitude between two time intervals is not defined.
- Independent variable is discrete time 'n' and the signal is represented by $x[n]$.

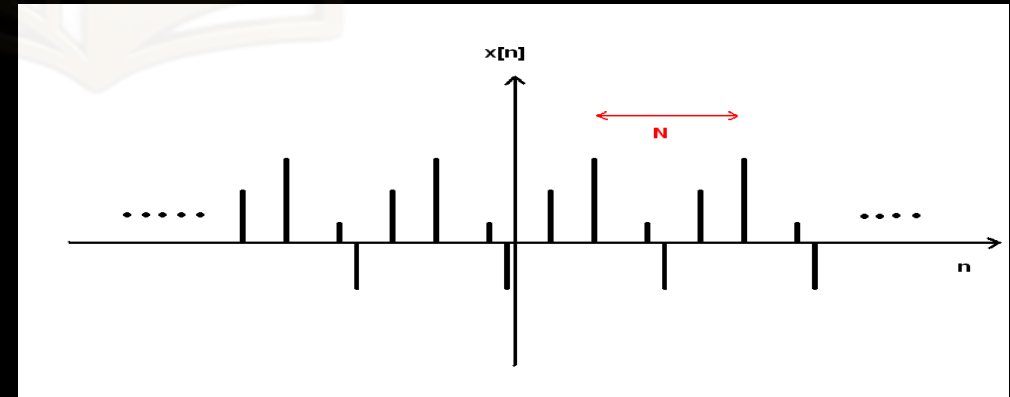
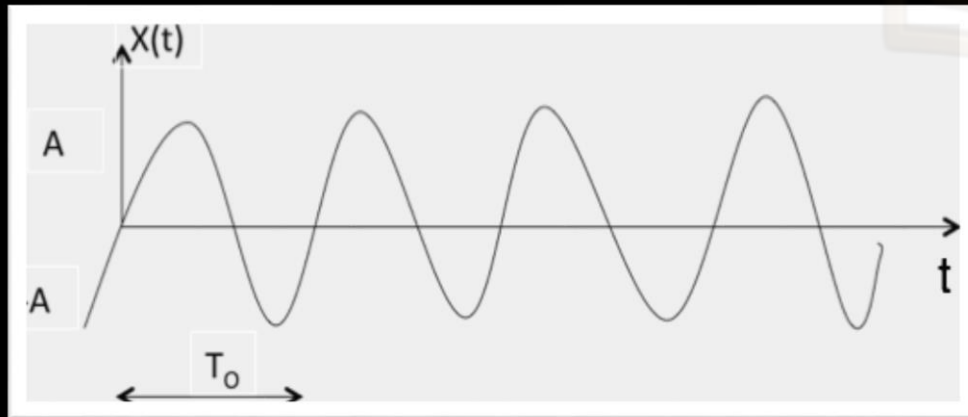


- A *discrete-time signal* is a sequence $x[n]$ defined for all integers $-\infty < n < \infty$.

Periodic and aperiodic signal:

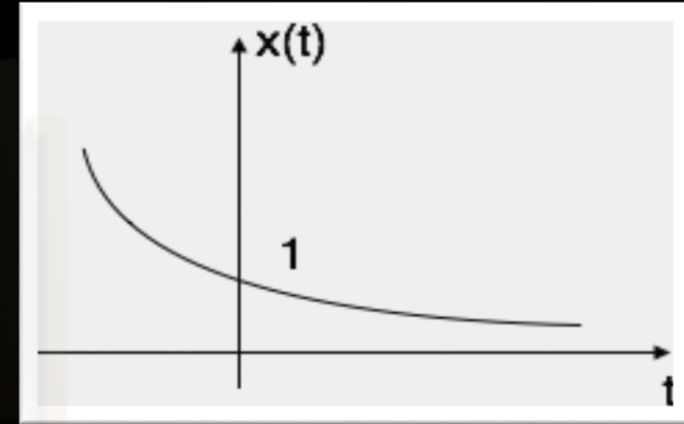
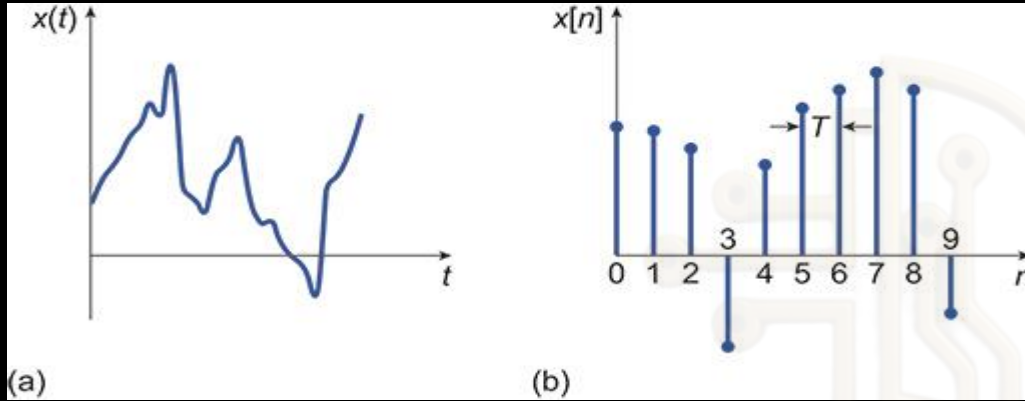
Periodic signal:

- Signal that has a definite pattern and repeats over a certain period T
- For continuous time signal, it is periodic if $x(t+T)=x(t)$ for $-\infty < t < \infty$, where T is its period.
- For discrete time signal, it is periodic if $x[n+N]=x[n]$ for $-\infty < n < \infty$, where N is its sampling period.



Aperiodic signal:

- Signal is said to be aperiodic if it does not repeat.
- They are said to have time period equal to infinity.



Deterministic and non deterministic signals:

Deterministic signals:

- Signals can be completely specified in time.
- Pattern of these signals are regular and can be characterised mathematically.
- At any time, nature and amplitude can be predicted.

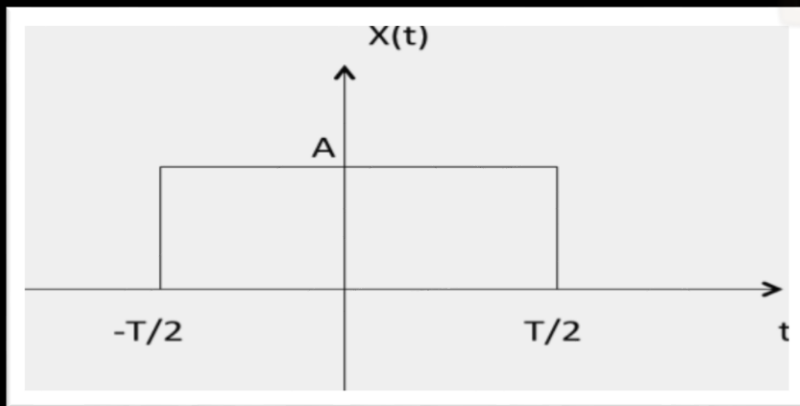
Non deterministic signals:

- Occurrence is always random in nature.
- Pattern is quite irregular.
- Also known as random or stochastic signal.
- Eg: thermal noise generated in electric ckt.

Even and odd signal:

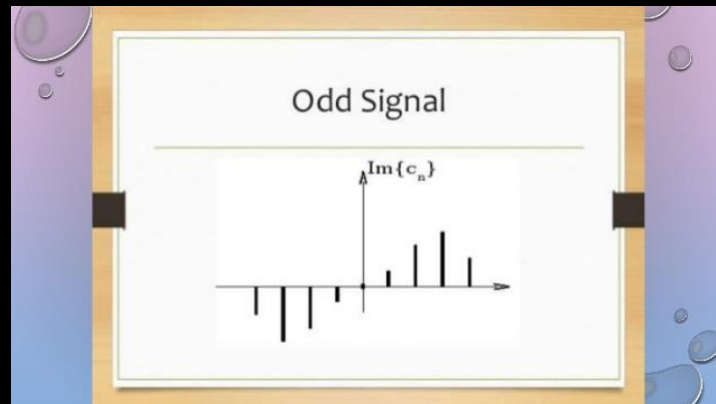
Even signal:

- A signal $x(t)$ or $x[n]$ is said to be even if
$$x(t) = x(-t) \text{ for continuous time signal}$$
$$x[n] = x[-n] \text{ for discrete time signal.}$$
- These signals are identical about the origin and exhibit symmetry in the time domain.



Odd signal:

- A signal is said to be odd if:
 $x(t) = -x(-t)$ for continuous time signal
 $x[n] = -x[-n]$ for discrete time signal.
- These signals exhibit anti symmetry and is not identical about the origin.
- The signal is identical to its negative. Eg: Sine wave .



Energy and power signal:

Energy signal:

- Energy signal has finite energy and zero signal power.
- $x(t)$ is an energy signal if $0 < E < \infty$ and $P = 0$.
- Eg: single rectangular pulse.
- The *total energy* of a continuous-time signal $x(t)$, where $x(t)$ is defined for $-\infty < t < \infty$, is

$$E_x \triangleq \lim_{T \rightarrow \infty} \int_{-T}^T |x(t)|^2 dt = \int_{-\infty}^{+\infty} |x(t)|^2 dt,$$

Power signal:

- Power signal has finite average power and infinite energy.
- $X(t)$ is a power signal if $0 < P < \infty$ and $E = \infty$.
- Eg: rectangular pulse train
- The power of a signal is calculated as:

$$P_{\infty} \triangleq \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt$$

Unit step function:

- A signal is said to be unit step if

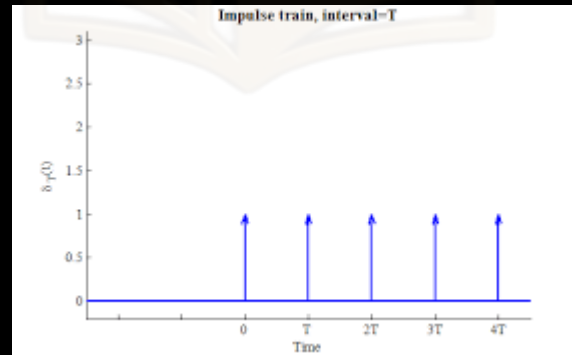
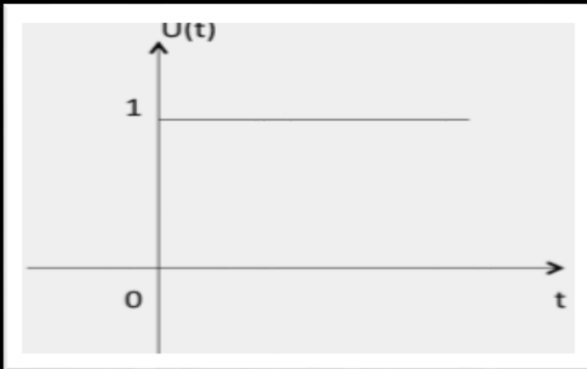
$$u(t) = 0 \text{ for } t < 0$$

$$= 1 \text{ for } t \geq 0 \quad \text{for continuous time signal.}$$

- For discrete time signal,

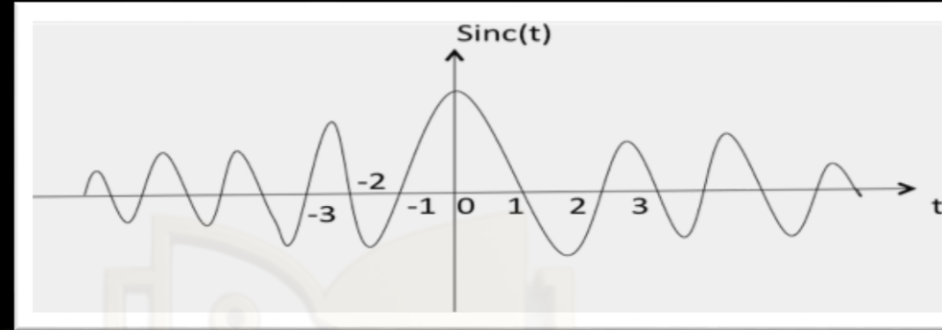
$$u[n] = 0 \text{ for } n < 0$$

$$= 1 \text{ for } n \geq 0$$



Sinc function:

- A signal is said to be a Sinc function if $x(t) = \text{Sinc}(t) = \text{Sint}/t$.



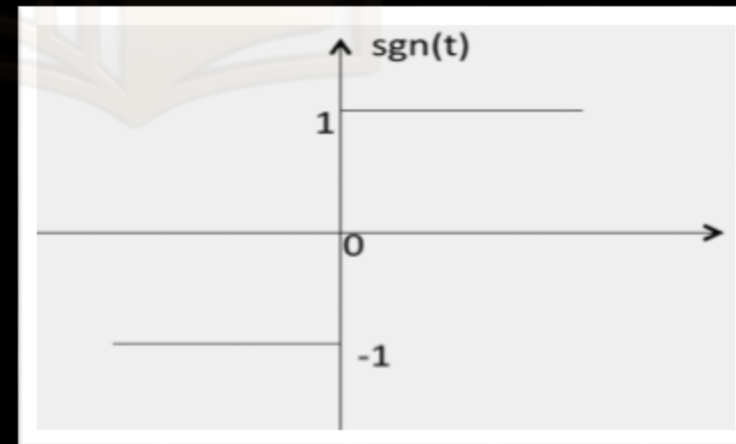
It is the inverse fourier transform of rectangular function.

Signum function:

$$X(t) = 1 \text{ for } t > 0$$

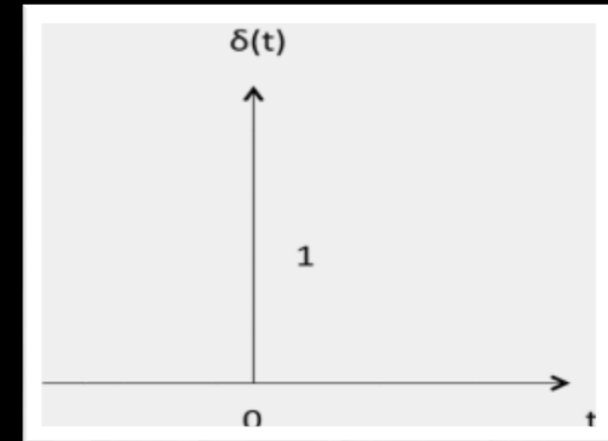
$$= -1 \text{ for } t < 0$$

$$= 0 \text{ for } t = 0$$

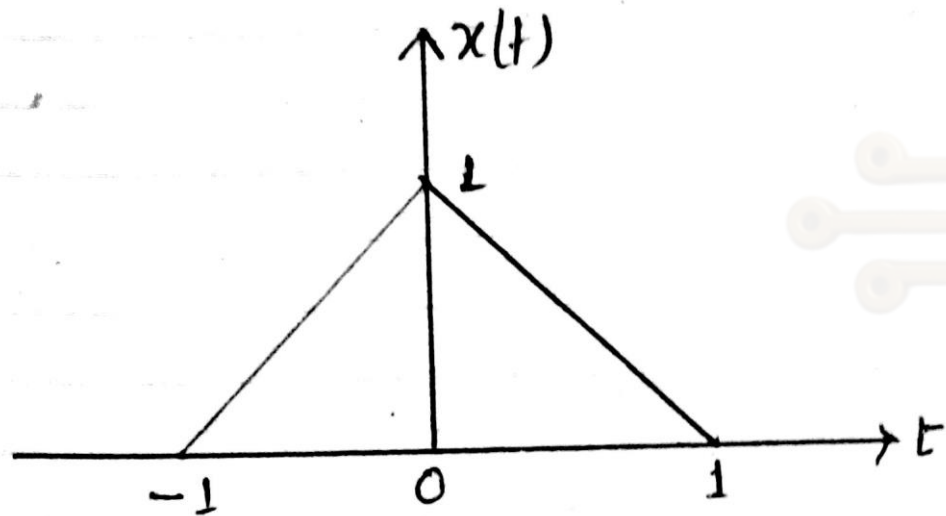


Delta function/ Impulse function

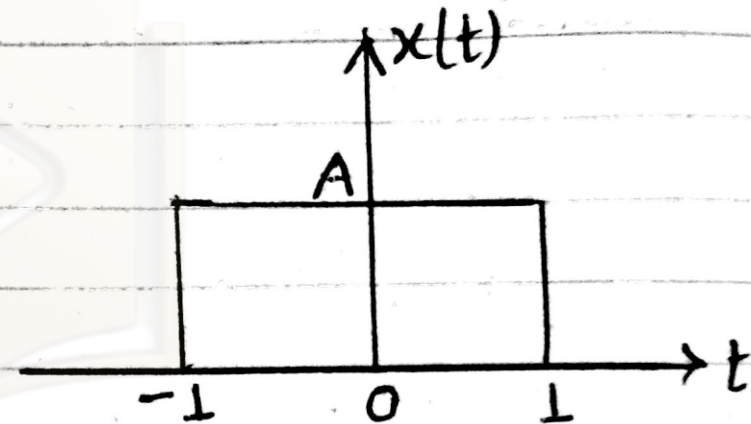
- Signal is said to be a delta function if $\delta(t) = 0$ for $t \neq 0$.



Triangular and rectangular pulses



$$x(t) = 1 - |t| \text{ for } |t| < 1$$
$$= 0 \text{ otherwise.}$$



$$x(t) = A \text{ for } |t| \leq 1$$
$$= 0 \text{ for } |t| > 1$$